Quantum Safe Cryptography for Post Quantum Era

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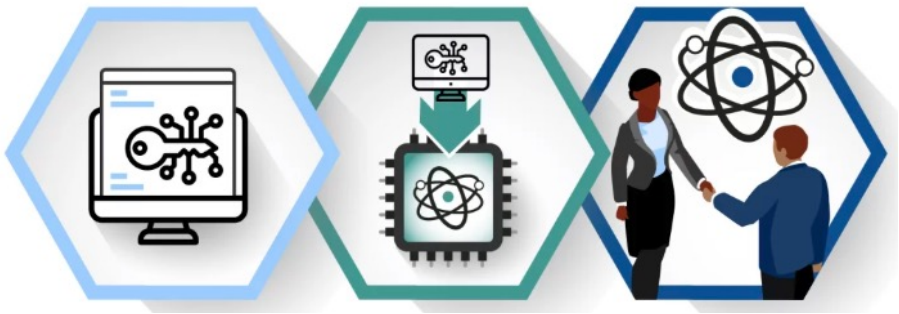
Why Quantum Pose a Threat to Existing Cryptography

- Can break current encryption methods that project sensitive data
- Making data vulnerable to theft and decryption
- Specific risks:
  - Breaking public-key cryptography
  - Compromising asymmetric encryption
  - Enabling “Harvest Now, Decrypt Later Attacks”
  - Critical Infrastructure Security
  - Impacting Supply Chains



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Preparing for Security in Post Quantum Era

- NIST initiated a process to solicit, evaluate, and standardize one or more quantum-resistant public-key cryptographic algorithms.

- The Open Quantum Safe (OQS) project is an open-source project that aims to support the transition to quantum-resistant cryptography. OQS is part of the Linux Foundation's Post-Quantum Cryptography Alliance.

OPEN QUANTUM SAFE

software for the transition to quantum-resistant cryptography

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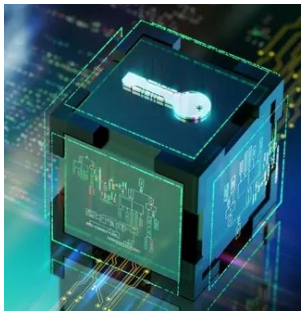
Quantum-safe Cryptography



Impact of Quantum on :

- Digital Signatures
- Key-establishment
- Symmetric Cryptography
- Application specific standards and guidelines

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Aim of this projects

Enabling the students to understand:

- Get familiarity with libraries and software stack to be able to implement quantum safe cryptography
- Understanding some of the algorithms and how they can be implemented
- Deploying post quantum security in secure software stack

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